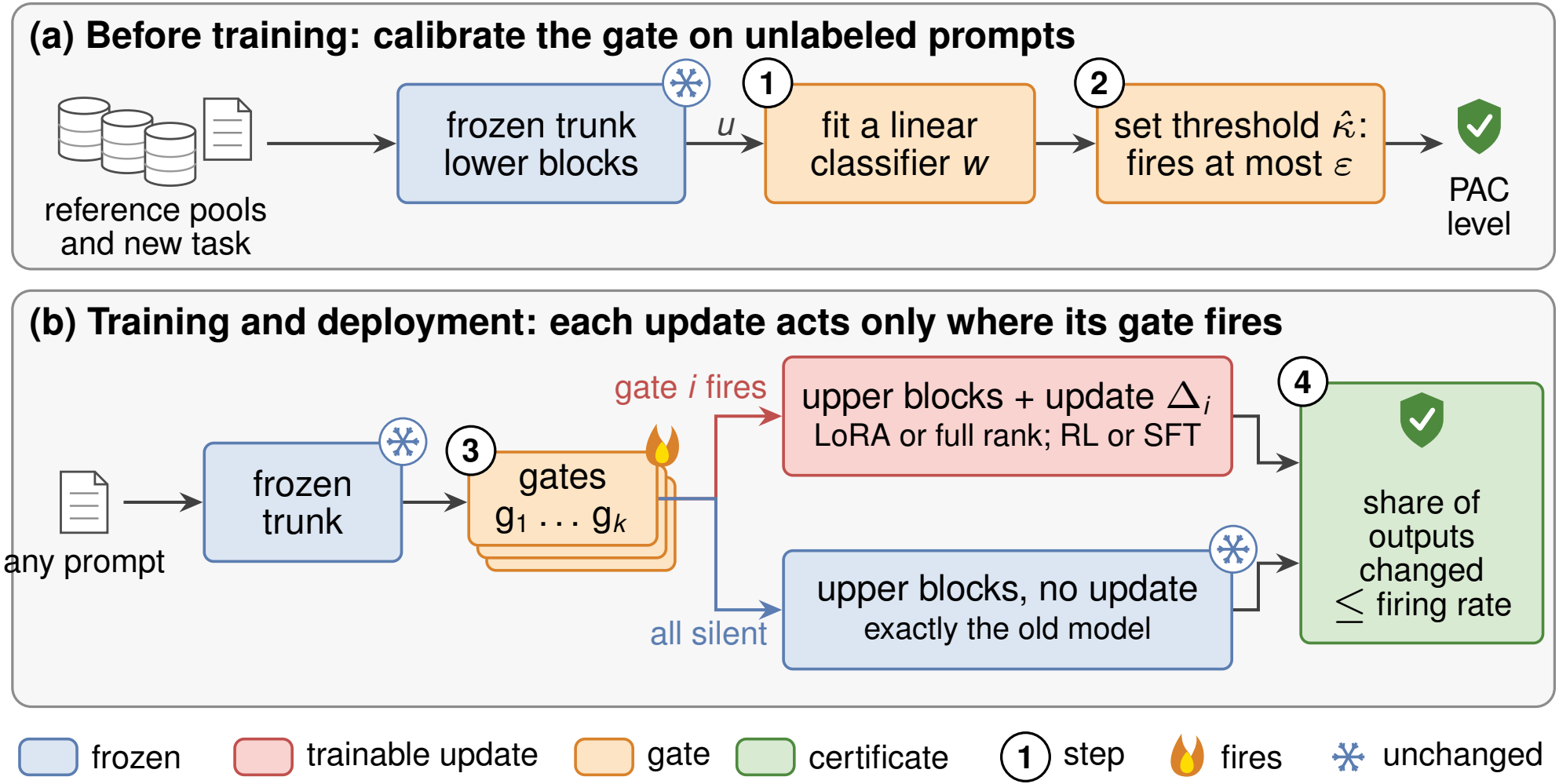
(a) Before training: calibrate the gate on unlabeled prompts
reference pools and new task
frozen trunk lower blocks
u
1
fit a linear classifier w
2
set threshold κ̂: fires at most ε
PAC level
(b) Training and deployment: each update acts only where its gate fires
any prompt
frozen trunk
3
gates g1 . . . gk
gate i fires
upper blocks + update Δi
LoRA or full rank; RL or SFT
all silent
upper blocks, no update
exactly the old model
4
share of outputs changed ≤ firing rate
frozen
trainable update
gate
certificate
1
step
fires
unchanged